

Matrix – Determinant – System of Equations

Matrix

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- 2 Elementary Row Operations and Rank of a Matrix
- 3 Matrix Operations.
- 4 Inverse of matrices.
- 5 Praticice



Consider the system of equations

$$\begin{cases} 2x + y - z = 2 \\ x - 2y + 2z = 1 \\ 3x - y + z = 3 \end{cases}$$



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$$\begin{cases} 2x + y - z = 2 \\ x - 2y + 2z = 1 \\ 3x - y + z = 3 \end{cases}$$

We can rewrite it in a simpler form

$$\begin{pmatrix} 2 & 1 & -1 & 2 \\ 1 & -2 & 2 & 1 \\ 3 & -1 & 1 & 3 \end{pmatrix}.$$

A 3×4 **matrix** is a numerical table consisting of 3 rows and 4 columns. Each number in the matrix is called an **element** or **entry**.



To solve the system of equations, we use the elimination method

$$\begin{pmatrix} 2 & 1 & -1 & 2 \\ 1 & -2 & 2 & 1 \\ 3 & -1 & 1 & 3 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -2 & 2 & 1 \\ 0 & 5 & -5 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

A row echelon matrix has 2 non-zero rows $\Rightarrow$ Rank=2.



Terminology	Definition
Leading entry	The first non-zero element in a row of a matrix.
(Row) echelon matrix	The reduced system.
Elementary Row Operations	Matrix equivalence transformation.
Rank	The actual number of equations in a system.



- 1 **Matrix Concepts**
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Definition of a matrix

An $m \times n$ **matrix** is a rectangular table of numbers containing m of rows and n of columns

$$A = \begin{pmatrix} a_{11} & \dots & a_{1j} & \dots & a_{1n} \\ \dots & \dots & \dots & \dots & \dots \\ a_{i1} & \dots & a_{ij} & \dots & a_{in} \\ \dots & \dots & \dots & \dots & \dots \\ a_{m1} & \dots & a_{mj} & \dots & a_{mn} \end{pmatrix} = (a_{ij})_{m \times n}.$$

a_{ij} is called the entry at row i , column j .

The set of all $m \times n$ matrices is denoted by $M_{m \times n}(K)$, where K is usually the field of real numbers ($\mathbb{R}$) or complex numbers $\mathbb{C}$.



Example 1.

Consider two matrices

$$A = \begin{pmatrix} 3 & 4 & 1 \\ 2 & 0 & 5 \end{pmatrix}_{2 \times 3}, B = \begin{bmatrix} 1+i & 2 \\ 3-i & 4i \end{bmatrix}$$



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$A \in M_{2 \times 3}(\mathbb{R})$. Entries of a matrix A :

$$a_{11} = 3, a_{12} = 4, a_{13} = 1, a_{21} = 2, a_{22} = 0, a_{23} = 5.$$

Matrix $B \in M_{2 \times 2}(\mathbb{C})$ consists of complex entries: $b_{21} = 3 - i$



Zero matrix

all entries are 0

$$A = (a_{ij})_{m \times n} \quad s.t. \quad a_{ij} = 0, \forall i, j.$$

Ex. Zero matrix of size 2×3

$$0_{2 \times 3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$



Leading entry

The first nonzero element of a row from the left is called the **leading entry** of this row.

$$\begin{pmatrix} 0 & 1 & 3 & -1 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Row 3 consists entirely of zeros, so it has no leading entry.



Echelon form of a matrix

An **echelon form** of a matrix satisfies

- i. All nonzero rows are above any rows of all zeros.
- ii. Each leading entry of a row is in a column to the right of the leading entry of the row above it.
- iii. All entries in a column below a leading entry are zeros.

$$A = \begin{pmatrix} 2 & 1 & 0 & 0 & 2 \\ 0 & 0 & 3 & 2 & 0 \\ 0 & 0 & 0 & 0 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \text{ is in echelon form}$$



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$$B = \begin{pmatrix} 2 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \text{ is NOT in echelon form.}$$



$$C = \begin{pmatrix} -2 & 1 & 0 & -1 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & -3 \end{pmatrix} \text{ is NOT in echelon form}$$

$$D = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -4 \end{pmatrix} \text{ is NOT in echelon form}$$



Reduced echelon form

If a matrix in echelon form satisfies the following additional conditions, then it is in **reduced echelon form**:

- i. The leading entry in each nonzero row is 1.
- ii. Each leading 1 is the only nonzero entry in its column.

$$A = \begin{pmatrix} 1 & 5 & 0 & 3 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \text{ is a reduced echelon matrix}$$



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$$A = \begin{pmatrix} 1 & 5 & 0 & 3 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \text{ is a reduced echelon matrix}$$

$$B = \begin{pmatrix} 0 & 1 & 7 & 0 & -1 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \text{ is NOT a reduced echelon matrix.}$$



The Transpose of a Matrix

Given an $m \times n$ matrix A , the transpose of A is the $n \times m$ matrix, denoted by A^T , whose columns are formed from the corresponding rows of A .

The transpose of $A = (a_{ij})_{m \times n}$ is the matrix

$$A^T = (a'_{ij})_{n \times m} \quad \text{s.t.} \quad a'_{ij} = a_{ji}$$

—→ *Interchange rows and columns*



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—→ *Interchange rows and columns*

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 3 \end{pmatrix} \implies A^T = \begin{pmatrix} 1 & 2 \\ 2 & 0 \\ 3 & 3 \end{pmatrix}$$



Square matrix

a **square matrix** is a matrix with the same number of rows and columns: $M_n(K)$. n is called the order of matrix A .

The entries $a_{11}, a_{22}, \dots, a_{nn}$ are called the **diagonal elements** of a square matrix and they form the **main diagonal** of A .

Trace of a matrix is the sum of the diagonal entries

$$\text{trace}(A) = \text{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}.$$



$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & -2 & 0 \\ 0 & 2 & -3 & 2 \\ -1 & 1 & 2 & 0 \end{pmatrix}$$

The diagonal elements of A are $1, 1, -3, 0$.

The trace of A is: $\text{tr}(A) = 1 + 1 - 3 + 0 = -1$.



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The trace of $B = \begin{pmatrix} -1 & 3 & 0 \\ 2 & 1+i & 3-i \\ 2 & 1 & 1-2i \end{pmatrix}$ is



$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & -2 & 0 \\ 0 & 2 & -3 & 2 \\ -1 & 1 & 2 & 0 \end{pmatrix}$$

The diagonal elements of A are $1, 1, -3, 0$.

The trace of A is: $\text{tr}(A) = 1 + 1 - 3 + 0 = -1$.

The trace of $B = \begin{pmatrix} -1 & 3 & 0 \\ 2 & 1+i & 3-i \\ 2 & 1 & 1-2i \end{pmatrix}$ is $\text{trace}(B) = -1 + 1 + i + 1 - 2i = 1 - i$.



Triangular matrix

- Matrix $A = (a_{ij})_n$ is called **upper triangular matrix** iff $a_{ij} = 0, \forall i > j$.
→ *all elements below the diagonal entries are zero.*
- Matrix $A = (a_{ij})_n$ is called **lower triangular matrix** iff $a_{ij} = 0, \forall i < j$.
→ *all elements above the diagonal entries are zero.*
- **Diagonal matrix** is a square matrix with all non-diagonal elements equal to zero.
→ only the diagonal entries of the square matrix can be nonzero.

$$a_{ij} = 0, \forall i \neq j.$$

- **Identity matrix** is a diagonal matrix with diagonal elements equal to one.
It is often denoted by I_n .

$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 2 & 0 \\ 0 & 0 & -2 \end{pmatrix}$ is an upper triangular matrix.

$B = \begin{pmatrix} 1 & 0 & 0 \\ -3 & 0 & 0 \\ 3 & 2 & -2 \end{pmatrix}$ is a lower triangular matrix.

$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ is a diagonal matrix

$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ is an identity matrix of order 3. Identity matrix of order 2:

$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$



symmetric matrix and skew-symmetric matrix

- $A \in M_n$ is called a symmetric matrix iff $A^T = A$.
- $A \in M_n$ is called a skew-symmetric matrix iff $A^T = -A$.

$$M = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & -3 \\ 2 & -3 & 4 \end{pmatrix} \text{ is symmetric.}$$

$$N = \begin{pmatrix} 0 & -1 & 2 \\ 1 & 0 & -3 \\ -2 & 3 & 0 \end{pmatrix} \text{ is skew-symmetric.}$$



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Elementary Row Operations

- **Scaling** Multiply all entries in a row by a nonzero constant $\alpha \neq 0$

$$r_i := \alpha r_i, \quad \alpha \neq 0$$

- **Replacement** Replace one row by the sum of itself and a multiple of another row

$$r_i := r_i + \beta r_j, \quad \forall \beta$$

- **Interchange** Swap two rows

$$r_i \leftrightarrow r_j$$

Matrix equivalence

$$A \xrightarrow{\text{elementary row operations}} B$$

Matrix B is equivalent to matrix A



Theorem

Every matrix is row equivalent to an echelon matrix.

A matrix may be row reduced into more than one matrix in echelon form.

Theorem

Each matrix is row equivalent to **one and only one** reduced echelon matrix.



Definition

- If a matrix A is row equivalent to an echelon matrix B , we call B an echelon form (or a row echelon form) of A .
- If B is in reduced echelon form, we call B the reduced echelon form of A .
- A **pivot position** in a matrix A is a location in A that corresponds to a leading 1 in its reduced echelon form.
- A **pivot column** is a column of A that contains a pivot position.



Example.

Apply elementary row operations to transform the following matrix first into **echelon form** and then into **reduced echelon form**

$$A = \begin{pmatrix} 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{pmatrix}$$



Step 1. Begin with the leftmost nonzero column

$$A = \begin{pmatrix} 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{pmatrix}$$

Step2. Select a nonzero entry in the pivot column as a pivot.

$$A \xrightarrow{r_1 \leftrightarrow r_3} \begin{pmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{pmatrix}$$



Step 3. Use row replacement operations to create zeros in all positions below the pivot.

$$\xrightarrow{r_2 := r_2 - r_1} \begin{pmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{pmatrix}$$

Step 4. Cover (or ignore) the row containing the pivot position and cover all rows, if any, above it. Repeat steps 1 to 3 until there are no more nonzero rows to modify.

$$\xrightarrow{r_3 := r_3 - \frac{3}{2}r_2} \begin{pmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{pmatrix} = B$$



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$$\xrightarrow{r_3 := r_3 - \frac{3}{2}r_2} \begin{pmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{pmatrix} = B$$

B is an echelon form of A .

Steps 1–4 is called the **forward phase** of the row reduction algorithm.



Step 5. Beginning with the rightmost pivot and working upward and to the left, create zeros above each pivot. If a pivot is not 1, make it 1 by a scaling operation.

$$B \xrightarrow{r_2 := r_2 - 2r_3} \begin{pmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 0 & -14 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{pmatrix} \xrightarrow{r_1 := r_1 - 6r_3} \begin{pmatrix} 3 & -9 & 12 & -9 & 0 & -9 \\ 0 & 2 & -4 & 4 & 0 & -14 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{pmatrix}$$

The next pivot is in row 2.

$$\xrightarrow{r_2 := \frac{1}{2}r_2} \begin{pmatrix} 3 & -9 & 12 & -9 & 0 & -9 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{pmatrix} \xrightarrow{r_1 := r_1 + 9r_2} \begin{pmatrix} 3 & 0 & -6 & 9 & 0 & -72 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{pmatrix}$$



Finally, scale row 1,

$$\begin{pmatrix} 3 & 0 & -6 & 9 & 0 & -72 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{pmatrix} \xrightarrow{r_1 := \frac{1}{3} r_1} \begin{pmatrix} 1 & 0 & -2 & 3 & 0 & -24 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{pmatrix} = C$$

C is an reduced echelon form of A .

Step 5 is called the **backward phase**, which produces the unique **reduced echelon form**.



Finally, scale row 1,

$$\begin{pmatrix} 3 & 0 & -6 & 9 & 0 & -72 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{pmatrix} \xrightarrow{r_1 := \frac{1}{3} r_1} \begin{pmatrix} 1 & 0 & -2 & 3 & 0 & -24 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{pmatrix} = C$$

C is an reduced echelon form of A .

Step 5 is called the **backward phase**, which produces the unique **reduced echelon form**.

Numerical note: Partial pivoting strategy: In step 2, select as a pivot the entry in a column having the largest absolute value to reduce roundoff errors in the calculations.



Rank of a matrix

Denoted as $\text{rank}(A)$ or $r(A)$

$r(A)$ = number of nonzero rows in its reduced echelon form .

Properties.

Let $A \in M_{m \times n}$

- i) $r(A) = r(A^T)$
- ii) $r(A) \leq \min\{m, n\}$
- iii) Elementary row operations do NOT change rank of matrices.



Example 2.

Find *Rank* of matrices $A = \begin{pmatrix} 1 & 1 & -1 & 2 & 1 \\ 2 & 3 & -1 & 4 & 5 \\ 3 & 2 & -3 & 7 & 4 \\ -1 & 1 & 2 & -3 & 1 \end{pmatrix}$. $B = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 2 & -1 & -4 & 3 \\ -1 & 1 & 3 & 0 \\ -1 & 1 & 3 & -2 \end{pmatrix}$.

$$C = \begin{pmatrix} -1 & 1 & 2 & 1 \\ 2 & 1 & 4 & -1 \\ -4 & 1 & 0 & m \\ -1 & 0 & 1 & 3 \end{pmatrix} . D = \begin{pmatrix} 1 & 2 & 1 & -1 \\ 2 & 2 & 3 & 2 \\ -1 & 3 & -1 & 4 \\ 3 & 1 & 0 & m \end{pmatrix} . E = \begin{pmatrix} m & 1 & 1 \\ 1 & m & 1 \\ 1 & 1 & m \end{pmatrix} .$$



Example 2.

Find *rank*

$$A = \begin{pmatrix} 1 & 1 & -1 & 2 & 1 \\ 2 & 3 & -1 & 4 & 5 \\ 3 & 2 & -3 & 7 & 4 \\ -1 & 1 & 2 & -3 & 1 \end{pmatrix}$$



Example 2.

Find *rank*

$$A = \begin{pmatrix} 1 & 1 & -1 & 2 & 1 \\ 2 & 3 & -1 & 4 & 5 \\ 3 & 2 & -3 & 7 & 4 \\ -1 & 1 & 2 & -3 & 1 \end{pmatrix} \xrightarrow{\substack{r_2 := r_2 - 2r_1 \\ r_3 := r_3 - 3r_1 \\ r_4 := r_4 + r_1}} \begin{pmatrix} 1 & 1 & -1 & 2 & 1 \\ 0 & 1 & 1 & 0 & 3 \\ 0 & -1 & 0 & 1 & 1 \\ 0 & 2 & 1 & -1 & 2 \end{pmatrix}$$

$$\xrightarrow{\substack{r_3 := r_3 + r_2 \\ r_4 := r_4 - 2r_2}} \begin{pmatrix} 1 & 1 & -1 & 2 & 1 \\ 0 & 1 & 1 & 0 & 3 \\ 0 & 0 & 1 & 1 & 4 \\ 0 & 0 & -1 & -1 & -4 \end{pmatrix} \xrightarrow{r_4 := r_4 + r_3} \begin{pmatrix} 1 & 1 & -1 & 2 & 1 \\ 0 & 1 & 1 & 0 & 3 \\ 0 & 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\implies r(A) = 3.$$



$$B = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 2 & -1 & -4 & 3 \\ -1 & 1 & 3 & 0 \\ -1 & 1 & 3 & -2 \end{pmatrix}.$$



$$B = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 2 & -1 & -4 & 3 \\ -1 & 1 & 3 & 0 \\ -1 & 1 & 3 & -2 \end{pmatrix}.$$

$$C = \begin{pmatrix} -1 & 1 & 2 & 1 \\ 2 & 1 & 4 & -1 \\ -4 & 1 & 0 & m \\ -1 & 0 & 1 & 3 \end{pmatrix}$$



$$B = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 2 & -1 & -4 & 3 \\ -1 & 1 & 3 & 0 \\ -1 & 1 & 3 & -2 \end{pmatrix}.$$

$$C = \begin{pmatrix} -1 & 1 & 2 & 1 \\ 2 & 1 & 4 & -1 \\ -4 & 1 & 0 & m \\ -1 & 0 & 1 & 3 \end{pmatrix}$$

$$r(B) = 3$$

$$r(C) = 3 \iff m = 3.$$



Example 2

$$D = \begin{pmatrix} 1 & 2 & 1 & -1 \\ 2 & 2 & 3 & 2 \\ -1 & 3 & -1 & 4 \\ 3 & 1 & 0 & m \end{pmatrix}$$



Example 2

$$D = \begin{pmatrix} 1 & 2 & 1 & -1 \\ 2 & 2 & 3 & 2 \\ -1 & 3 & -1 & 4 \\ 3 & 1 & 0 & m \end{pmatrix}$$

$$E = \begin{pmatrix} m & 1 & 1 \\ 1 & m & 1 \\ 1 & 1 & m \end{pmatrix}$$



Example 2

$$D = \begin{pmatrix} 1 & 2 & 1 & -1 \\ 2 & 2 & 3 & 2 \\ -1 & 3 & -1 & 4 \\ 3 & 1 & 0 & m \end{pmatrix}$$

$$E = \begin{pmatrix} m & 1 & 1 \\ 1 & m & 1 \\ 1 & 1 & m \end{pmatrix}$$

$$r(D) = 3 \iff m = \frac{-108}{5}.$$



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Addition and Scalar Multiplication

Let $A, B \in M_{m \times n}$ (same order) and a scalar $\alpha \in K$.

Addition $A + B = (a_{ij} + b_{ij})$: Addition of corresponding elements.

Multiplication $\alpha.A = (\alpha.a_{ij})$: Multiply α by all the elements of A .

Subtraction $A - B = A + (-1).B$.

Definition Two matrices A and B are equal if their sizes are the same and $a_{ij} = b_{ij}, \forall i, j$.



Properties of Addition and Scalar Multiplication

- i. $A + B = B + A.$
- ii. $(A + B) + C = A + (B + C).$
- iii. $A + 0 = A.$
- iv. $\alpha(A + B) = \alpha A + \alpha B.$
- v. $\alpha(\beta A) = (\alpha\beta)A.$
- vi. $(\alpha + \beta)A = \alpha A + \beta A.$



Example 4.

Compute

a. $\begin{pmatrix} 1 & 2 & -1 \\ 2 & -1 & 0 \end{pmatrix} + \begin{pmatrix} 3 & -2 & 1 \\ 1 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 4 & 0 & 0 \\ 3 & -1 & 3 \end{pmatrix}.$

b. $2 \cdot \begin{pmatrix} 1 & 2 & -1 \\ 2 & -1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 4 & -2 \\ 4 & -2 & 0 \end{pmatrix}.$



Example 4.

Compute

a. $\begin{pmatrix} 1 & 2 & -1 \\ 2 & -1 & 0 \end{pmatrix} + \begin{pmatrix} 3 & -2 & 1 \\ 1 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 4 & 0 & 0 \\ 3 & -1 & 3 \end{pmatrix}.$

b. $2 \cdot \begin{pmatrix} 1 & 2 & -1 \\ 2 & -1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 4 & -2 \\ 4 & -2 & 0 \end{pmatrix}.$

c. $2 \begin{pmatrix} 1 & 2 & -1 \\ 2 & -1 & 0 \end{pmatrix} - 3 \begin{pmatrix} 3 & -2 & 1 \\ 1 & 0 & 3 \end{pmatrix} = \begin{pmatrix} -7 & 10 & -5 \\ 1 & -2 & -9 \end{pmatrix}$

d. $3 \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & -1 \end{pmatrix} - 4 \begin{pmatrix} -1 & 0 \\ 2 & 1 \\ -3 & 6 \end{pmatrix}$ is not defined (because they are not the same size).



Matrix Multiplication

Let $A = (a_{ij})_{m \times p}$, $B = (b_{ij})_{p \times n}$.

The multiplication $A.B = C = (c_{ij})_{m \times n}$

$$\begin{pmatrix} \dots & \dots & \dots & \dots \\ a_{i1} & a_{i2} & \dots & a_{ip} \\ \dots & \dots & \dots & \dots \end{pmatrix} \cdot \begin{pmatrix} \dots & b_{1j} & \dots \\ \dots & b_{2j} & \dots \\ \dots & \dots & \dots \\ \dots & b_{pj} & \dots \end{pmatrix} = \begin{pmatrix} \dots & \dots & \dots & \dots \\ \dots & c_{ij} & \dots \\ \dots & \dots & \dots \end{pmatrix}.$$

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ip}b_{pj}, \quad i = \overline{1, m}, j = \overline{1, n}.$$



Matrix Multiplication

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The multiplication $A.B = C = (c_{ij})_{m \times n}$

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$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{ip}b_{pj}, \quad i = \overline{1, m}, j = \overline{1, n}.$$

Require : *The number of columns in the first matrix equals the number of rows in the second matrix*

Rule:

$$c_{ij} = r_i(A) \times c_j(B)$$



Example 5.

Let $A = \begin{pmatrix} 2 & -1 & 4 \\ 4 & 1 & 0 \end{pmatrix}$; $B = \begin{pmatrix} 1 & -2 & 2 \\ 3 & 0 & 1 \\ 2 & 4 & 3 \end{pmatrix}$. Compute $B \cdot A$, $A \cdot B$, $I \cdot A$, $A \cdot I$.



Example 5.

Let $A = \begin{pmatrix} 2 & -1 & 4 \\ 4 & 1 & 0 \end{pmatrix}$; $B = \begin{pmatrix} 1 & -2 & 2 \\ 3 & 0 & 1 \\ 2 & 4 & 3 \end{pmatrix}$. Compute $B \cdot A$, $A \cdot B$, $I \cdot A$, $A \cdot I$.

a) BA : is NOT defined.

b) $AB = \begin{pmatrix} 2 & -1 & 4 \\ 4 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & -2 & 2 \\ 3 & 0 & 1 \\ 2 & 4 & 3 \end{pmatrix} = \begin{pmatrix} 7 & 12 & 15 \\ 7 & -8 & 9 \end{pmatrix}.$

c) $I_2 \cdot A =$



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c) $I_2 \cdot A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 & 4 \\ 4 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & -1 & 4 \\ 4 & 1 & 0 \end{pmatrix}$



Example 5.

Let $A = \begin{pmatrix} 2 & -1 & 4 \\ 4 & 1 & 0 \end{pmatrix}$; $B = \begin{pmatrix} 1 & -2 & 2 \\ 3 & 0 & 1 \\ 2 & 4 & 3 \end{pmatrix}$. Compute $B \cdot A$, $A \cdot B$, $I \cdot A$, $A \cdot I$.

a) BA : is NOT defined.

b) $AB = \begin{pmatrix} 2 & -1 & 4 \\ 4 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & -2 & 2 \\ 3 & 0 & 1 \\ 2 & 4 & 3 \end{pmatrix} = \begin{pmatrix} 7 & 12 & 15 \\ 7 & -8 & 9 \end{pmatrix}.$

c) $I_2 \cdot A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 & 4 \\ 4 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & -1 & 4 \\ 4 & 1 & 0 \end{pmatrix}$

d) $A \cdot I_3 = \begin{pmatrix} 2 & -1 & 4 \\ 4 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -1 & 4 \\ 4 & 1 & 0 \end{pmatrix}$



Properties of Multiplication

Let A be an $m \times n$ matrix, and let B, C have sizes for which the indicated sums and products are defined.

- i. $A(BC) = (AB)C$. (associative law)
- ii. $A(B + C) = AB + AC$. (left distributive law)
- iii. $(B + C)A = BA + CA$. (right distributive law)
- iv. $I_m A = A I_n = A$. (identity)
- v. $\alpha(AB) = (\alpha A)B = A(\alpha B)$, for any scalar α
- vi. $(AB)^T = B^T A^T$
- vii. $\text{trace}(AB) = \text{trace}(BA)$
- viii. For any $x \in \mathbb{R}^n$, $A(Bx) = (AB)x$



Remarks

- i. In general, $AB \neq BA$ (does not commute).
- ii. The cancellation laws do NOT hold for matrix multiplication, i.e., if $AB = AC$, then it is NOT true in general that $B = C$.
- iii. If a product AB is the zero matrix, you canNOT conclude in general that either $A = 0$ or $B = 0$.



Power of Matrix

Let $A \in M_n$. Denote: $A^0 = I_n$

$A^m = A.A...A.A$ (The multiplication of m copies of matrix A), $m \in \mathbb{Z}_+^*$.



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Matrix Polynomial

Let $A \in M_n$ and a polynomial $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_mx^m$. Then

$$f(A) = a_0I_n + a_1A + a_2A^2 + \cdots + a_mA^m.$$



Example 6.

Let $A = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}$. a/ $f(x) = 2x^2 - 4x + 3$. Compute $f(A)$.



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$$\text{a/ } f(A) = 2A^2 - 4A + 3I_2.$$

$$= 2 \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}^2 - 4 \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix} + 3 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} =$$

$$2 \begin{pmatrix} 1 & -6 \\ 18 & 13 \end{pmatrix} - \begin{pmatrix} 8 & -4 \\ 12 & 16 \end{pmatrix} + \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} -3 & -8 \\ 24 & 13 \end{pmatrix}$$



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b/ Let $f(x) = x^2 - 6x + 11$. Compute $f(A)$.



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b/ Let $f(x) = x^2 - 6x + 11$. Compute $f(A)$.

$$f(A) = A^2 - 6A + 11.I_2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$



Example 7.

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & -1 \end{pmatrix}$, $B = \begin{pmatrix} -1 & 0 \\ 2 & 1 \\ 3 & 2 \end{pmatrix}$, $C = \begin{pmatrix} -1 & 2 \\ 3 & 4 \end{pmatrix}$. Compute

- a. $A + 2B; 3A - 2B^T$
- b. AB, BA, ABC, BAC
- c. $A.A^T, A^T.A$



Example 7.

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & -1 \end{pmatrix}$, $B = \begin{pmatrix} -1 & 0 \\ 2 & 1 \\ 3 & 2 \end{pmatrix}$, $C = \begin{pmatrix} -1 & 2 \\ 3 & 4 \end{pmatrix}$. Compute

- a. $A + 2B; 3A - 2B^T$
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- c. $A.A^T, A^T.A$
- d. $C^8, f(C)$ where $f(x) = x^2 - 3x - 10$.



Example 7.

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- 1 Matrix Concepts
- 2 Elementary Row Operations and Rank of a Matrix
- 3 Matrix Operations.
- 4 Inverse of matrices.**
- 5 Praticce



Inverse of matrices

An $n \times n$ matrix A is said to be invertible if there is an $n \times n$ matrix B such that:

$$AB = BA = I.$$

In this case, B is the **inverse** of A , denoted as A^{-1} . In fact, B is uniquely determined by A .



Example 8.

Let $A = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$ và $B = \begin{pmatrix} -3 & 2 \\ 2 & -1 \end{pmatrix}$



Example 8.

Let $A = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$ và $B = \begin{pmatrix} -3 & 2 \\ 2 & -1 \end{pmatrix}$

We have $AB = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} -3 & 2 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$$BA = \begin{pmatrix} -3 & 2 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

thus B is the inverse of A and conversely: $A^{-1} = B$.



Properties of Inverse Matrices

Let 2 $n \times n$ matrices A, B , invertible

- i) A^{-1} is unique.
- ii) If A is an invertible matrix, then A^{-1} is invertible and $(A^{-1})^{-1} = A$
- iii) If A is an invertible matrix, so is A^T and $(A^T)^{-1} = (A^{-1})^T$.
- iv) If A and B are $n \times n$ invertible matrices, then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$



Theorem

An $n \times n$ matrix A is invertible if and only if A is row equivalent to I_n , and in this case, any sequence of elementary row operations that reduces A to I_n also transforms I_n into A^{-1} .

Algorithm for Finding $[I|A^{-1}]$

Row reduce the augmented matrix $[A|I]$. If A is row equivalent to I , then $[I|A^{-1}]$. Otherwise, A does not have an inverse.

$$[A|I] \xrightarrow{\text{elementary row operations}} [I|A^{-1}]$$



Example 9.

Find the inverse matrix of $A = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$.



Example 9.

Find the inverse matrix of $A = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$.

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 2 & 5 & 0 & 1 \end{array} \right] \xrightarrow{r_2 - 2r_1} \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{array} \right]$$

$$\xrightarrow{r_1 - 2r_2} \left[\begin{array}{cc|cc} 1 & 0 & 5 & -2 \\ 0 & 1 & -2 & 1 \end{array} \right]$$

thus A is invertible and

$$\Rightarrow A^{-1} = \begin{pmatrix} 5 & -2 \\ -2 & 1 \end{pmatrix}.$$



Example 10.

Find the inverse matrix of $B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{pmatrix}$



Example 10.

Find the inverse matrix of $B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{pmatrix}$

$$[B|I] = \left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 2 & 0 & 1 & 0 \\ 1 & 2 & 3 & 0 & 0 & 1 \end{array} \right) \rightarrow \left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 & 1 \end{array} \right)$$

$$\xrightarrow[r_1 - r_2]{r_3 - r_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & -1 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{array} \right) \xrightarrow{r_2 - r_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & -1 & 0 \\ 0 & 1 & 0 & -1 & 2 & -1 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{array} \right)$$

thus B is invertible and $B^{-1} = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}$.



Example 11.

Find the inverse matrix of $A = \begin{pmatrix} 2 & 1 \\ -1 & 3 \end{pmatrix}$ $B = \begin{pmatrix} 0 & 2 & 1 \\ -1 & 1 & 2 \\ 3 & 1 & -1 \end{pmatrix}$



Example 12.

Let $A = \begin{pmatrix} 1 & 2 \\ -1 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 3 & 1 \\ -1 & 5 \end{pmatrix}$. Find X so that

a/ $AX = 3B^T + 2X$ b/ $XA + 3B = A^T + 2X$

c/ $A(2X + B) = 3B^T - X$ d/ $AX - XB = 3A^T$



Example 12.

Let $A = \begin{pmatrix} 1 & 2 \\ -1 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 3 & 1 \\ -1 & 5 \end{pmatrix}$. Find X so that

a/ $AX = 3B^T + 2X$ b/ $XA + 3B = A^T + 2X$

c/ $A(2X + B) = 3B^T - X$ d/ $AX - XB = 3A^T$



Example 13.

Let $A = \begin{pmatrix} 0 & 1 \\ 3 & -1 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. Find matrix X satisfies

a/ $2AX = B^T - 3X$ b/ $2XA - 4B^T = 2A^T + 3X$

c/ $A(3X + B^T) = B - 2X$ d/ $AX + XB = 2B^T$



Summary

General matrix concepts

- Matrix is
- The transpose of matrix A is.....
- The leading entry of a row.....
- Row echelon form.....
- Rank of a matrix.....



Summary

Key concepts related to square matrices

- Square matrix...
- Trace of a matrix...
- Symmetric matrix...
- Skew-symmetric matrix...
- Upper triangular matrix...
- Lower triangular matrix ...
- Diagonal matrix....

Summary

Matrix operations

- Multiply a scalar α with a matrix.
- Addition: requirement..... compute.....
- Multiplicationn
 requirement.....
 compute.....
- Power of matrix.....

Let $f(x) = 2x^3 + 6x - 7 \Rightarrow f(A) = \dots\dots\dots$

Summary

Elementary row operations

- Elementary row operations

O1:.....

O2:.....

O3:.....

- Rank of a matrix



Summary

The inverse of a matrix

- The condition for a matrix to be invertible.....
- Properties of an inverse matrix....

$$(A^{-1})^{-1} = \quad (AB)^{-1} =$$

$$A \text{ is invertible then } r(AB) = \dots \quad r(BA) = \dots$$

- Algorithm to find the inverse of matrix

.....



- 1 Matrix Concepts
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Example 14: Find Rank

$$A = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 2 & -1 & -4 & 3 \\ -1 & 1 & 3 & 0 \end{pmatrix}, B = \begin{pmatrix} 1 & 2 & 1 & 1 \\ 5 & 10 & 5 & 6 \\ 3 & 6 & 3 & 4 \end{pmatrix}.$$

$$C = \begin{pmatrix} 1 & 1 & -1 & 2 & 1 \\ 1 & -1 & -2 & 3 & -1 \\ 3 & 2 & -3 & 7 & 4 \\ -1 & 1 & 2 & -3 & 1 \end{pmatrix}$$



Example 15: find the rank of a matrix depending on m

$$D = \begin{pmatrix} 1 & 2 & 1 & 1 \\ 3 & 3 & 4 & 0 \\ 1 & 5 & 0 & m \end{pmatrix}, E = \begin{pmatrix} 1 & 1 & 2 & 1 \\ 2 & 2 & m & 0 \\ 1 & 2 & 0 & m-2 \\ 2 & 3 & 1 & 0 \end{pmatrix}, F = \begin{pmatrix} m & 1 & 1 & 1 \\ 1 & m & 1 & 1 \\ 1 & 1 & m & 1 \\ 1 & 1 & 1 & m \end{pmatrix}.$$



Example 16.

$$\text{Let } A = \begin{pmatrix} 1 & 1 & 2 & -1 \\ 2 & 1 & 1 & 3 \\ 3 & 1 & 2 & -2 \end{pmatrix}, B = \begin{pmatrix} 2 & 0 & 1 \\ 3 & 1 & -1 \\ -1 & 0 & 3 \\ -2 & 1 & 0 \end{pmatrix}, C = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

a/ Compute (if applicable): $AB, BA, ABC, ACB, A^2, A.A^T$.

b/ Find Trace of $AB, BA, AA^T, B^T B$.



Example 17.

Let $A = \begin{pmatrix} -1 & 2 \\ 3 & 4 \end{pmatrix}$ và $f(x) = x^2 - 3x - 10$.

a/ Compute A^2, A^4, A^8 .

b/ Compute $f(A)$

c/ Compute A^{2021} .



Example 18.

Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} -1 & 0 \\ 2 & 1 \end{pmatrix}$. Find matrix X so that

a/ $2AX = B^T - 3X$ b/ $2XA - 4B^T = 2A^T + 3X$ c/ $A(3X + B^T) = B - 2X$

d/ $AX + XB = 2B^T$



Example 19. Find the inverse of matrix

$$A = \begin{pmatrix} 2 & -1 \\ 3 & 2 \end{pmatrix}$$



Example 19. Find the inverse of matrix

$$A = \begin{pmatrix} 2 & -1 \\ 3 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 1 & -1 \\ 2 & 3 & 1 \\ 3 & 4 & 1 \end{pmatrix}$$



Example 19. Find the inverse of matrix

$$A = \begin{pmatrix} 2 & -1 \\ 3 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 1 & -1 \\ 2 & 3 & 1 \\ 3 & 4 & 1 \end{pmatrix}, C = \begin{pmatrix} 2 & 1 & 1 \\ 3 & 2 & 1 \\ -1 & 0 & 2 \end{pmatrix}$$
$$D = \begin{pmatrix} 3 & 1 & 1 & 1 \\ 1 & 3 & 1 & 1 \\ 1 & 1 & 3 & 1 \\ 1 & 1 & 1 & 3 \end{pmatrix}$$



Example 20.

Let $A = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 3 \\ 1 & -2 \end{pmatrix}$. Find matrix X so that

a/ $3XA + 4A^T = 2B - 2X$ b/ $B(2X - A^T) = 3B^T + 3X$

c/ $AX - 2XB = 3A^T$.



Example 21. Let

$$A = \begin{pmatrix} 2 & 3 & 1 & 1 \\ -1 & 2 & 1 & 3 \\ 3 & 0 & 1 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ 2 & -1 \\ -3 & 1 \\ -1 & 1 \end{pmatrix}$$

- a/ Compute $A + 2B$, BA , AB (If applicable).
- b/ Compute $A.A^T$, $A^T.A$, $B.B^T$, $B^T.B$. What kind of matrices are these?



THANK YOU FOR ATTENTION

